ATLAS Repair and Re-Audit

Deterministic–Nondeterministic Interfaces, Version 0.2

Proof, mathematical scope, priority, and typesetting review • 20 October 2026

Verdict: PASS for the repaired, restricted finite-classical mathematics.
Still preliminary: This is *not* an established general interface theory of all independently specified deterministic and probabilistic theories. The stronger finite-resource result is fully proved in the repaired text, but an exhaustive literature priority audit has not been completed.

1. Materials and audit standard

Compared the original eight-page article *DNI v0.1* and prior *ATLAS Audit v0.1* with the repaired eight-page source and rendered PDF, *DNI v0.2 ATLAS Repaired*. Reconstructed the proofs independently, checked the finite-support and state-count arguments, performed exhaustive three- and four-state graph checks and exact rational witness verification, validated cited precedent for the intertwining identities, and inspected all rendered PDF pages. This is an ATLAS-style mathematical audit conducted in this session, not a claim of execution by any separately running research system.

2. Disposition of all major earlier findings

ID	Status	Repair and evidence
M1: Priority	RESOLVED	(S1)–(S2) explicitly identified with $\Lambda\Phi = I$, $\Lambda P = K\Lambda$; Rogers–Pitman (1981) and Alberts cited before the Markov-history theorem. No originality claim on those identities.
M2: Possibility	RESOLVED	Added (P1)–(P2), finite-support bridge, support-composition proposition, shift-path existence, and a finite-state minimum corollary. The distinction between nondeterminism and stochastic weights is maintained.
M3: Minimality	RESOLVED (restricted)	Defined $m_*(K)$ over finite hidden spaces and proved exact necessary and sufficient graph criterion and minimal cardinality. General categorical/semantic minimality deliberately remains open.
M4: Unlimited encoding	RESOLVED (restricted)	Added finite-resource obstruction, minimum hidden-state recurrence, fixed-implementation LP, and two contrasting examples. The freedom to choose f, R, E is explicit.

3. Independent mathematical verification

Repaired result	Result	Independent check
Stochastic DNI and Markov histories	PASS	E concentrated on readout fibres and $f_*E = EK$ suffice for conditional hidden distributions to renew at each observed step, giving the full Markov joint law. Established Rogers–Pitman specialization.

Repaired result	Result	Independent check
Finite support bridge	PASS	For finite nonnegative sums, positive support of $EK(y)$ equals the union of supports of $E(z)$ for all positive $K(y,z)$. Pushforward support under a deterministic map equals its image.
Compositional support	PASS	Positivity of a finite matrix product is equivalent to the existence of a positive summand.
Possibilistic path shift	PASS	Every legal tail $z, z_1, \ldots$ may be prepended by an allowed predecessor y; therefore $f[L(y)] = \bigcup_{z \in N(y)} L(z)$.
Positive-word lower bound	PASS	A fixed finite hidden initial state fixes one output word; different initial readouts occupy disjoint preparation fibres. Every positive Markov word must occur. Hence $\lvert X\rvert \geq B_n$.
Graph-path equivalence	PASS	Cyclic branching permits arbitrarily delayed exits and yields unbounded distinct prefixes; without cyclic branching every walk is forced after its first repeated vertex, so only finitely many infinite paths exist.
Exact minimal realization	PASS	Path-space shift gives a witness when finite; the word bound is tight at a length separating all finitely many distinct infinite paths. Thus $m_* = \lvert\mathsf{Path}_\infty\rvert = \sup_n B_n$.
Possibilistic minimum	PASS	(P2) lifts every admissible finite word inductively to a hidden starting state, giving the same lower bound; finite shift-path realization attains it.
Counting recurrence	PASS	Every cyclic vertex has one forced infinite continuation; every noncyclic vertex splits legal paths by first successor. Noncyclic induced subgraph is acyclic.
Four-state witness	PASS	All 12 intertwining coordinates and nine readout coordinates were independently checked using exact rational arithmetic. Four paths give the matching lower bound.
Fixed f, R LP	PASS	Normalization, readout fibres and the coordinate form of $f_*E = EK$ are linear in E; rational coefficients admit a rational feasible point when nonempty.
Finite-time TV stability	PASS	Reconstructed telescoping/one-step contraction bound $\Delta_{n+1} \leq \Delta_n + \varepsilon$; it is marginal-only as written.
Elementary joint-model criterion	PASS	In a disjoint extension of a common signature, models can be transported across an isomorphism of common reducts and glued without mixed axioms. No stronger amalgamation theorem claimed.

4. Independent computational checks (not proof substitutes)

1. **Exhaustive three-state support graphs:** 343 directed graphs with all three outdegrees nonzero were enumerated ($7^3 = 343$). Exactly 39 satisfied the no-branching-on-cycles condition and 304 failed. In every satisfying case, length-11 path-word counts had stabilized and matched the graph recurrence. Every failing case had strictly more positive length-11 words than length-2 words. This checks many instances but is not an exhaustive verification of arbitrary state counts.

2. **Independent four-state exhaustive check:** All $15^4 = 50{,}625$ four-vertex directed graphs with nonempty outdegree were separately enumerated. Cycle membership was computed using positive-length transitive closure rather than the depth-first check above. Exactly 904 satisfied the graph criterion and 49,721 failed. Each satisfying graph had stabilized path-word counts by length 12; each failing graph

had more length-12 words than length-4 words. This provides a second finite consistency check, not a proof of the general theorem.

3. **Four-hidden-state example:** The rational interface with $E(a) = \frac{1}{3}\delta_{w_1} + \frac{2}{3}\delta_{w_2}$, $E(b) = \delta_{w_3}$, $E(c) = \delta_{w_4}$ was checked entry-by-entry for (S1) and (S2).

4. **Typesetting:** The revised TeX compiles to eight letter-size PDF pages with no overfull boxes, LaTeX errors, or undefined references in the final pass. All eight pages were rendered and reviewed; two minor underfull paragraph diagnostics do not conceal mathematics.

5. Remaining qualifications and publication advice

Q1 — Originality: NOT CERTIFIED. The newly organized finite-state characterization is proved, but elementary path/cycle classification and automata realizations may have earlier formulations. The audit establishes mathematical correctness, not international literature priority. Present the result as a self-contained finite-classical theorem pending extended literature review.

Q2 — Theory-of-theories scope: LIMITED. The paper formally states a joint-model criterion for disjoint first-order signatures; it does not construct a universal mixed-signature interface or prove conservative embedding of arbitrary input theories. A syntactic pushout need not yield semantic compatibility.

Q3 — Theoretical costs: QUALIFIED. m_* optimizes the number of hidden states over freely chosen deterministic f, readout R, and preparation E. It does not minimize bits of storage, probability-description complexity, physical implementability, runtime, or a preassigned source deterministic theory.

Q4 — Semantics: LIMITED. Probability is classical and discrete time; no nonclassical measurement, contextuality, causal locality or physical reconstruction theorem is established. The role of statistical preparation remains declared, rather than suppressed.

Q5 — Algorithmic precision: ACCEPTABLE. The exact recurrence is finite and testable. Bit complexity and optimization when the source structures are independently fixed remain future work.

6. Final certification statement

Mathematical proofs: PASS for results within stated hypotheses.
Prior ATLAS M1–M4 repairs: PASS in the expressly restricted sense above.
Priority and broader philosophical/physical applicability: NOT CERTIFIED.
Document quality: PASS.
Recommended designation: *ATLAS-audited working mathematical article, Version 0.2.* The parent Version 0.1 remains unchanged; no publication action was taken.

References checked

1. L. C. G. Rogers and J. W. Pitman, "Markov functions," *Annals of Probability* **9**(4), 573–582 (1981). https://doi.org/10.1214/aop/1176994363.

2. T. Alberts, "Pitman–Rogers on Finite State Markov Chains," University of Utah, https://www.math.utah.edu/~alberts/notes/sle-markov-functions/discrete-pitman-rogers/.

3. T. Fritz, "A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics," *Advances in Mathematics* **370**, 107239 (2020).

4. J. A. Goguen and R. M. Burstall, "Institutions: Abstract model theory for specification and programming," *Journal of the ACM* **39**(1), 95–146 (1992).